

Shared-range accuracy: BWTandem is second to ULTRA under every overlap rule.

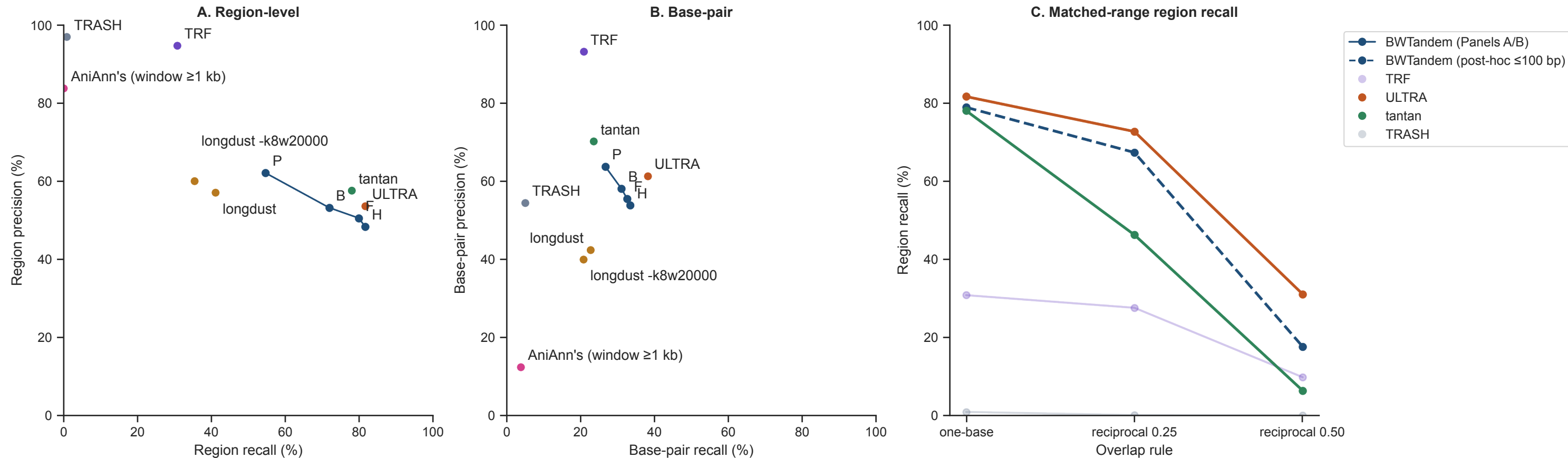


Figure 1. (A) Region-level and (B) base-pair precision–recall on GRCh38 against the adotto catalog (one-base overlap). BWTandem's four configurations (P, B, F, H) are connected lines; competing tools are single points. (C) Region recall evaluated only for periods ≤ 100 bp across three overlap rules. ULTRA performs best, with BWTandem second. BWTandem data in (C) uses full-range output filtered post-hoc to ≤ 100 bp to ensure accurate boundary assessment. Reciprocal rules here stress-test boundary precision against a merged catalog rather than absolute ranking. The 2026 tools are excluded from (C) as they lack period bounds.